

AI Developer Assignment:

Assignment Objective

Build a mini autonomous AI agent that can act as a “Newsletter Agent”.

What You Need to Build

Create a **single autonomous agent** with the following capabilities:

1. **Goal Input** The agent receives a plain English goal: "Create a weekly newsletter on latest AI agent news and send it to our subscribers."
2. **Agent Behavior (True Agentic)** The agent must autonomously:
 - Research the latest AI agent news (use any public API, search tool, or scraping you prefer).
 - Summarize the top 5-7 relevant articles.
 - Generate a clean newsletter (HTML or Markdown).
 - Simulate sending it (save as file or print the email content + subject).
3. **Core Requirements**
 - Use **multi-step reasoning** (planning → research → writing → review → output).
 - Show **tool use** (at least 2-3 tools: web search, summarizer, HTML generator, etc.).
 - Include **self-reflection / critique steps** (let the agent evaluate and improve its own output).
 - Make it **autonomous** — one function call like `run_newsletter_agent(goal)` should do the full job.
 - Add a simple toggle between Fully Autonomous and Human-in-the-Loop mode.
 - Clean, well-structured, readable code.

Tech Stack

- User LangChain or pure LangGraph
- Use Gemini, Grok, Claude, OpenAI, or local LLM (Ollama) - your choice.
- Simple front end to interact with the agent.